

University Notice Management System

Build a **complete production-ready Flutter application** named **University Notice Management System** with **Flutter (latest stable)** for frontend and **Laravel REST API + MySQL** for backend.

Core Objective

Create a secure centralized platform where universities can publish, manage, and distribute notices digitally. Users can register/login, view notices by category, receive alerts, and admins can manage notices through role-based dashboards.

Tech Stack

Frontend

- Flutter (Android + iOS + Web ready)
- Riverpod / Provider for state management
- Dio / HTTP for API integration
- GoRouter for navigation
- SharedPreferences / Secure Storage for auth tokens
- Responsive modern Material 3 UI

Backend

- Laravel latest version
- REST API with Sanctum / JWT Authentication
- MySQL database
- Role-based access control

User Roles

1. Student / User

- Register account
- Login / Logout
- Forgot password
- Update profile
- View notices
- Filter notices by:
 - Department
 - Category
 - Date
 - Priority
- Download PDF notices
- Bookmark notices
- Receive push notifications
- Search notices
- Dark mode

2. Admin

- Admin login
- Publish notices
- Edit / Delete notices
- Manage categories
- Upload PDF/image attachments
- Manage departments
- View analytics
- Approve notices before publish

3. Super Admin

- Super admin login
- Manage all admins
- Manage university settings

- Manage users
- Global reports
- Role assignment
- Access logs
- Backup system

Notice Categories

Create dynamic category management:

- Academic
- Examination
- Admission
- Scholarship
- Hostel
- Placement
- Events
- Circular
- Emergency
- Others

Admin can add/edit/delete categories.

Flutter Screens Required

Public Screens

1. Splash Screen
2. Onboarding
3. Login
4. Register
5. Forgot Password

User Screens

6. Home Dashboard
7. Notice Feed
8. Notice Detail
9. Search Notice
10. Categories Screen
11. Saved Notices
12. Notifications Screen
13. Profile Screen
14. Settings

Admin Screens

15. Admin Dashboard
16. Create Notice
17. Edit Notice
18. Category Management
19. Department Management
20. User Management

Super Admin Screens

21. Super Admin Dashboard
22. Admin Management
23. System Reports
24. App Settings

UI Requirements

Create beautiful modern UI:

- University theme colors
- Responsive for mobile/tablet/web
- Clean cards for notices

- Notice badges:
 - Urgent
 - New
 - Expiring Soon
- Drawer navigation
- Bottom navigation
- Smooth animations
- Charts in admin dashboard

Notice Card Must Show

- Title
- Category
- Department
- Publish date
- Short description
- Attachment icon
- Priority badge
- View count

Backend API Modules

Authentication APIs

POST /register

POST /login

POST /logout

POST /forgot-password

Notice APIs

GET /notices

GET /notices/{id}

POST /notices

PUT /notices/{id}

DELETE /notices/{id}

Category APIs

GET /categories

POST /categories

PUT /categories/{id}

DELETE /categories/{id}

User Management APIs

GET /users

PUT /users/{id}

DELETE /users/{id}

Admin APIs

GET /dashboard/stats

Database Tables

users

- id
- name
- email
- phone

- password
- role

notices

- id
- title
- description
- category_id
- department_id
- attachment
- priority
- published_by
- publish_date
- expiry_date
- views

categories

- id
- name

departments

- id
- name

bookmarks

- id
- user_id
- notice_id

Security Features

- JWT / Sanctum auth
- Role-based middleware
- SQL injection prevention
- File upload validation
- API rate limit
- Secure token storage in Flutter
- Admin access restrictions

Advanced Features

- Push notifications (Firebase)
- PDF viewer in app
- Export notices to PDF
- Analytics dashboard
- Multi-language support
- Offline cached notices
- Real-time updates
- Notice expiry auto hide

Folder Structure (Flutter)

```
lib/  
├── core/  
├── models/  
├── services/  
├── providers/  
├── screens/  
├── widgets/  
├── routes/  
└── main.dart
```


Deliverables

Generate full source code with:

1. Flutter frontend complete
2. Laravel backend complete
3. SQL database schema
4. API integration
5. Responsive UI
6. Clean code architecture
7. README setup guide

Git commands

```
git checkout milan-sharma
```

```
git add .
```

```
git commit -m 'Note of modifications'
```

```
git push -u origin milan-sharma
```